

IS883: Deploying Generative AI

Mohannad Elhamod

Welcome!

What is this class about?

ChatGPT... Absolute Terror!

A Meta AI security researcher said an OpenClaw agent ran amok on her inbox

Source: techcrunch.com, 23 Feb 2026 — retrieved 24 Aug 2026

The Washington Post

TECH Artificial Intelligence Help Desk Internet Culture Space Tech Policy

INNOVATIONS

ChatGPT took their jobs. Now they walk dogs and fix air conditioners.

Technology used to automate dirty and repetitive jobs. Now, artificial intelligence chatbots are coming after high-paying ones.

By Pranshu Verma and Gerrit De Vynck

June 2, 2023 at 6:00 a.m. EDT

NBC NEWS POLITICS U.S. NEWS WORLD LOCAL SPORTS BUSINESS TIPLINE SHOPPING HEALTH WATCH SUBSCRIBE

ARTIFICIAL INTELLIGENCE

OpenAI agents hacked Hugging Face in 700-strong swarm, tried to cover tracks, investigations find

Forbes

FORBES > INNOVATION > ENTERPRISE TECH

Will ChatGPT Put Data Analysts Out Of Work?

Bernard Marr Contributor

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tics Business Tech Science Newsletters Fight for democracy

'ChatGPT said I did not exist': how artists and writers are fighting back against AI

ChatGPT can do anything!

MECHANICAL ENGINEERING

ROBOT PERFORMS FIRST REALISTIC SURGERY WITHOUT HUMAN HELP

System trained on videos of surgeries performs like an expert surgeon

Jill Rosen / Jul 9, 2025

SLS SLS BLOGS LEGAL AGGREGATE

GPT-4 Passes the Bar Exam: What That Means for Artificial Intelligence Tools in the Legal Profession

April 19, 2023 | By Pablo Arredondo, Q&A with Sharon Driscoll and Monica Schreiber

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OpenAI says it cracked 90-year-old maths problem in 88 hours

6 days ago

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Kali Hays
Technology reporter

NBC NEWS LIVE UPDATES: MOROCCO EARTHQUAKE POLITICS U.S. NEWS WORLD BUSINESS HEALTH NBC NEWS TIPLINE WATCH NOW

CULTURE MATTERS

ChatGPT passes MBA exam given by a Wharton professor



The Velvet Sundown: The AI Band Controversy Explained

Berklee faculty explain what you need to know about the now-confirmed AI band with over 1 million streaming subscribers.

July 22, 2025

By Tara Bellucci

ChatGPT is amazing!



How ChatGPT Can Help You Ace Your Next Interview

 OLLMOO
7,453 followers

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Replace Grammarly Premium with OpenAI ChatGPT

How to use OpenAI's ChatGPT to replace Grammarly Premium

 Sung Kim · [Follow](#)
Published in [Geek Culture](#) · 9 min read · Dec 19, 2022

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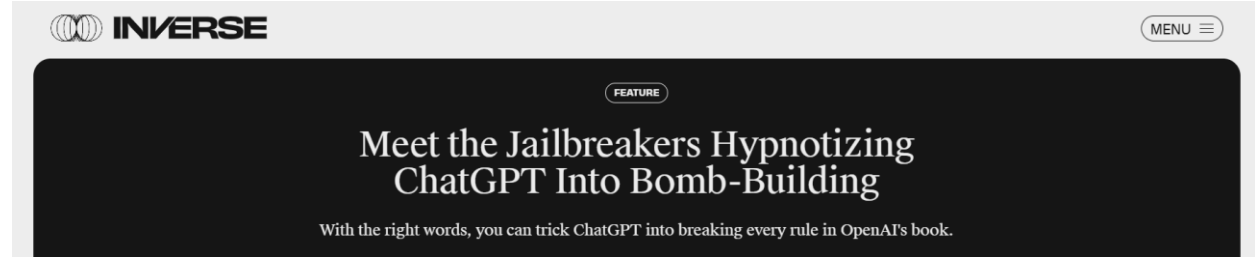
[HOME](#) > [TECH](#)

Programmers are pumped by the rise of ChatGPT, because it makes their jobs easier and helps people to find a lucrative career in tech

Emilia David Mar 3, 2023, 5:00 AM EST

ChatGPT is horrible!



ChatGPT.. What is it really about though?

Let's play a bit!

Is It Hype?

PRESS RELEASE

SEC Charges Two Investment Advisers with Making False and Misleading Statements About Their Use of Artificial Intelligence

Cruise confirms robotaxis rely on human assistance every four to five miles

PUBLISHED MON, NOV 6 2023·7:11 PM EST



Lora Kolodny
@IN/LORAKOLODNY/

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TRANSPORTATION

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Tesla engineer testifies that 2016 video promoting self-driving was faked

Rebecca Bellan

1:25 PM PST · January 17, 2023

Los Angeles Times

Nvidia's \$750-billion AI bet deepens fears of a circular tech bubble

Google faces controversy over edited Gemini AI demo video

PUBLISHED FRI, DEC 8 2023·1:20 PM EST | UPDATED FRI, DEC 8 2023·2:25 PM EST



Steve Kovach



Jennifer Elias

@IN/JENNIFER-ELIAS-845B1130/

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Are we on the verge of AGI?

1960s - The AI pioneer Marvin Minsky: “from three to eight years we will have a machine with the general intelligence of an average human being

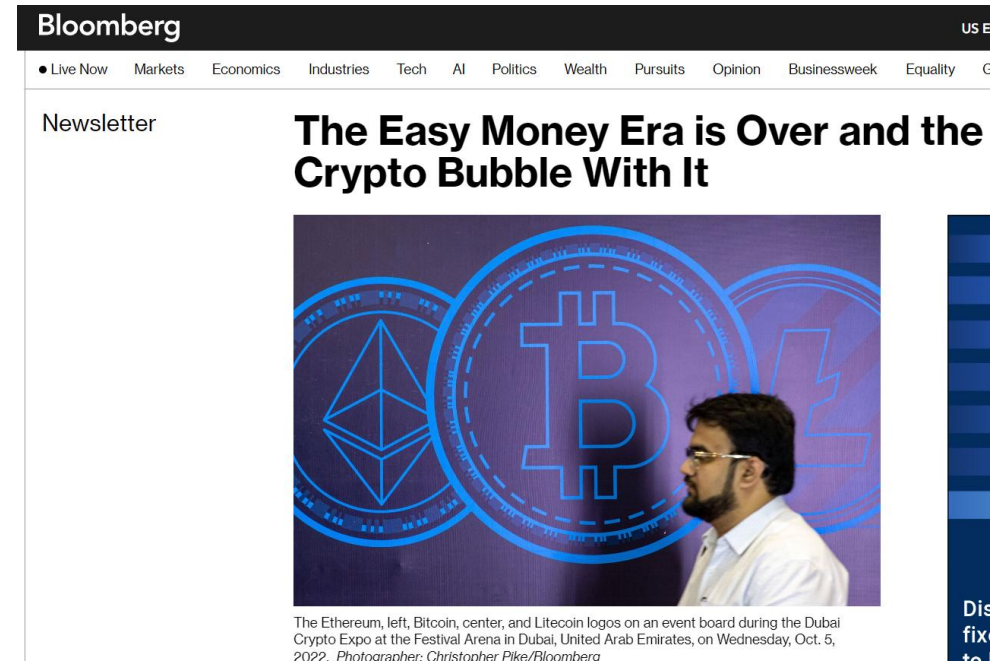
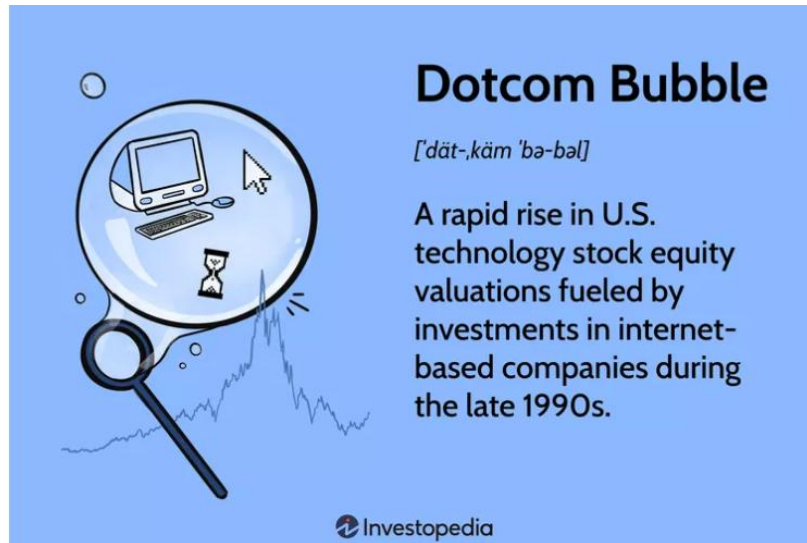
1980s - the successes of expert systems, *BusinessWeek* proclaimed “AI: it’s here.”

2020s - large language models, [Jensen Huang: "AGI has arrived. Congratulations @OpenAI team."](#)

...

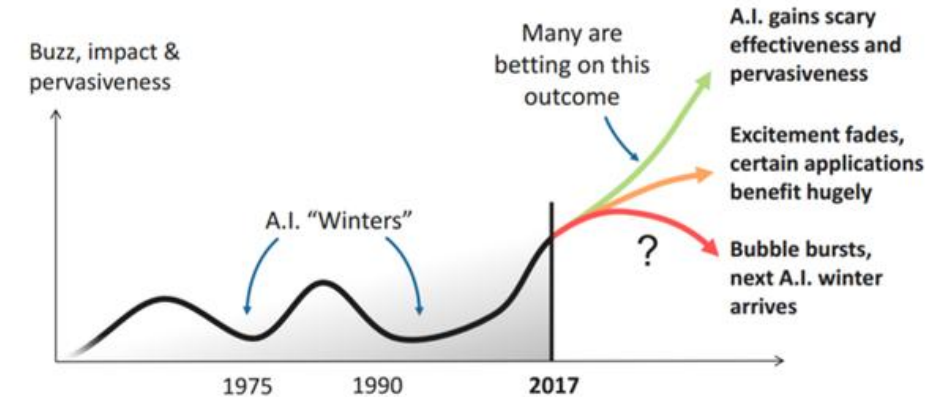
Max Bennett, A Brief History of Intelligence: Evolution, AI, and the Five Breakthroughs That Made Our Brains (HarperCollins, 2023), Introduction.

Is this a hype?



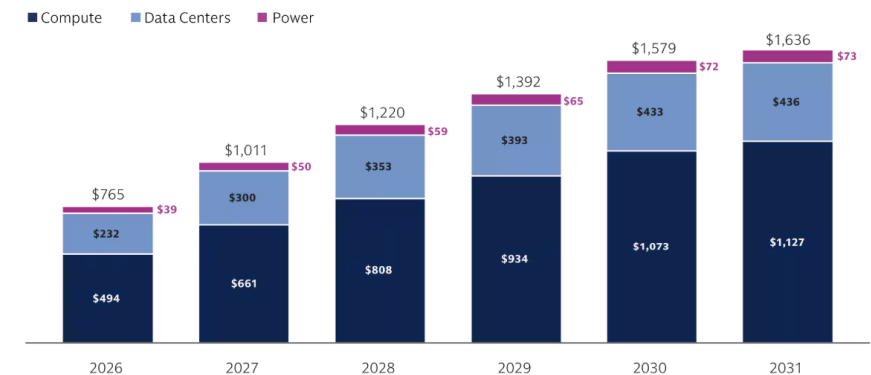
Is this a hype?

- Yann LeCun, one of the fathers of AI, believes the current technology (Transformers) is not the ultimate answer.
- Still, Transformers do seem to be getting extremely effective.
- Moreover, despite the effectiveness, the cost has been very high !



Baseline aggregate AI CapEx estimates (bn)

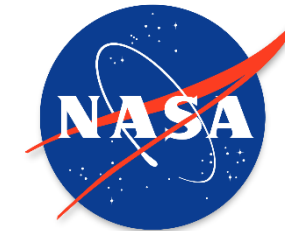
~\$7.6tr of capital between 2026 and 2031 across compute, data centers, and power



GoldmanSachs

General Guidelines

Who is your professor?



Why are you here?

- I am sure it eventually leads to money....
- Maybe it is just a required class?
- Nonetheless, understanding how something works is essential for gauging its potential and limitation.

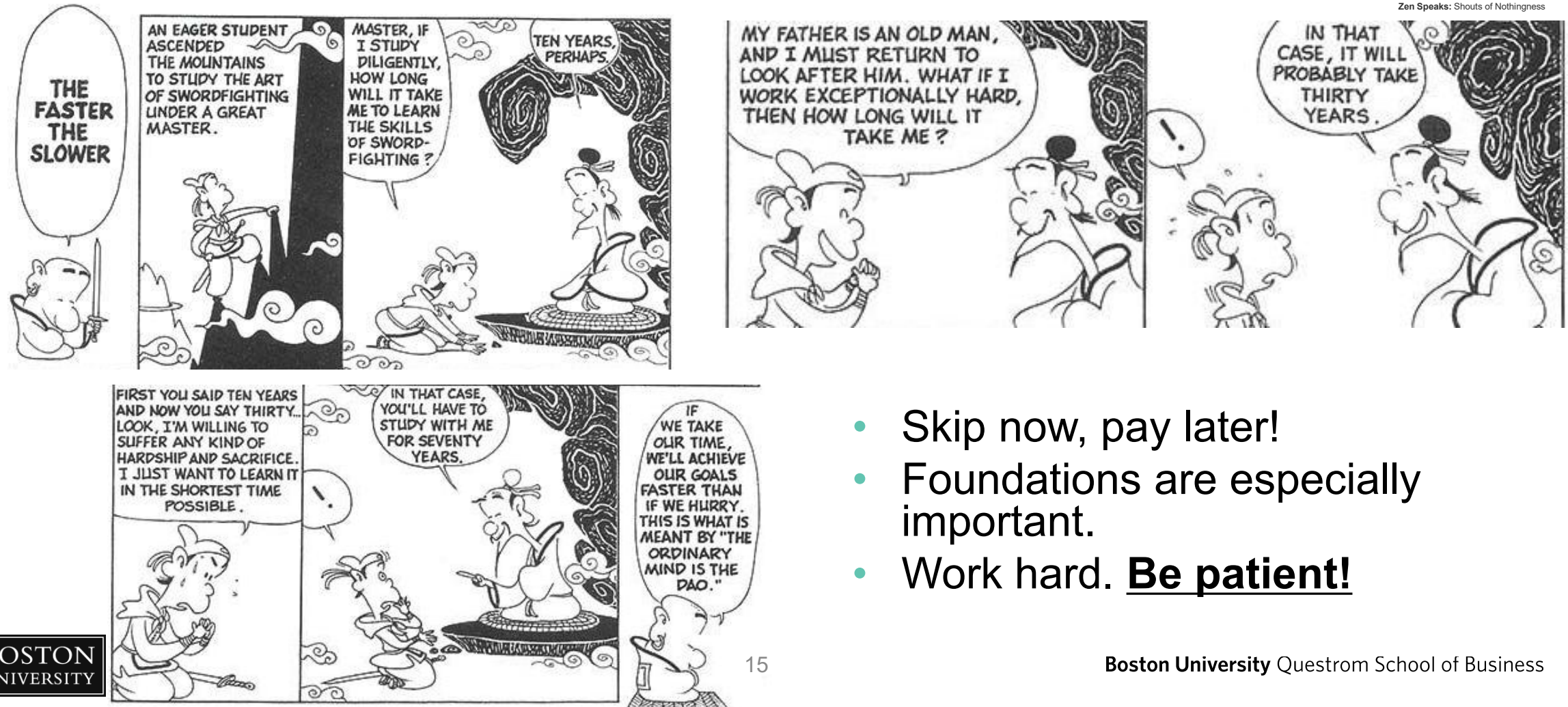


Euclid

Someone who had begun to [study] geometry asked Euclid, 'What shall I get by learning these things?' Euclid called his slave and said, 'Give him [some money], since he must make gain out of what he learns'.

(Heath, 1981, loc. 8625)

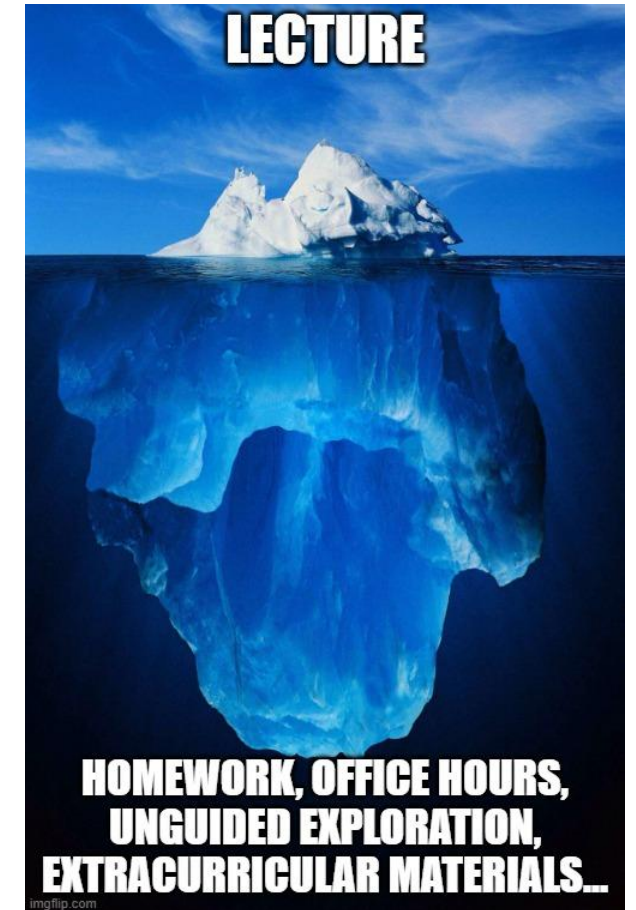
Fundamentals are important!



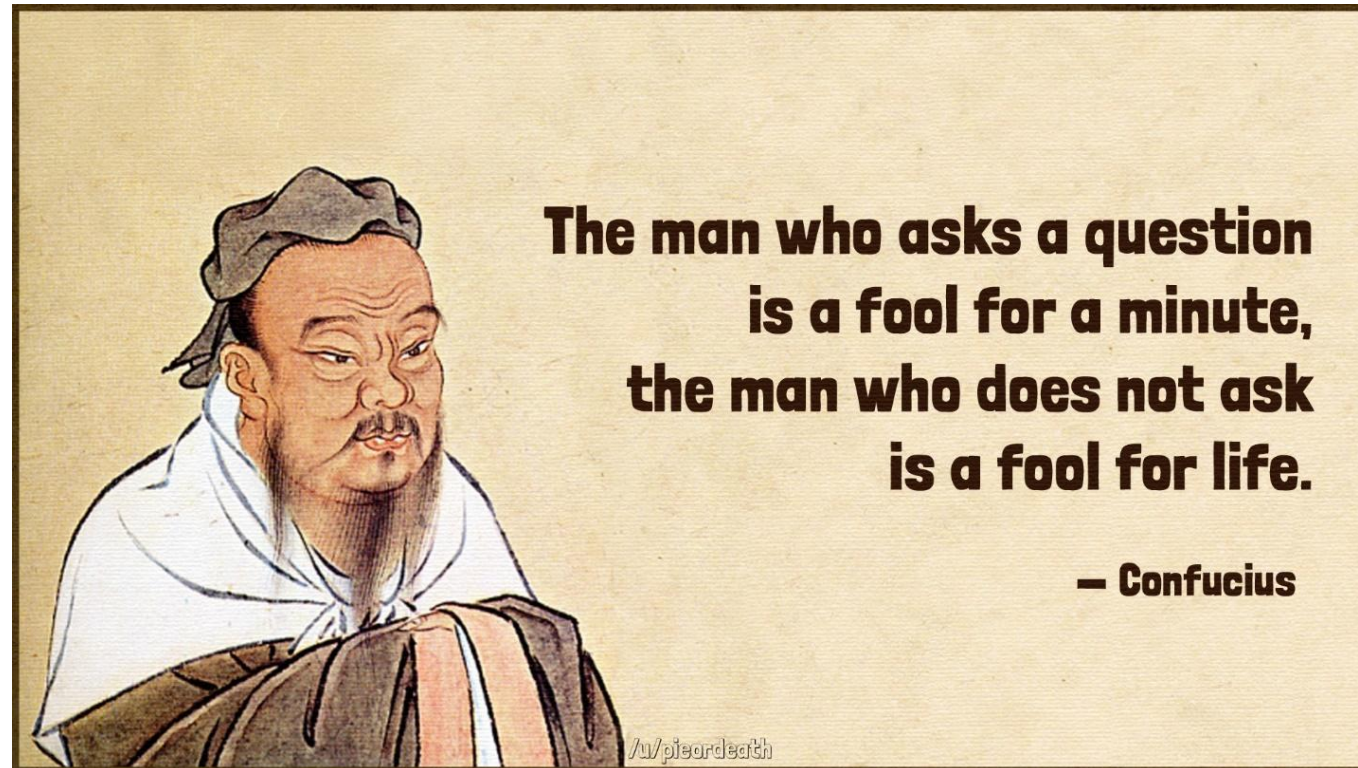
- Skip now, pay later!
- Foundations are especially important.
- Work hard. **Be patient!**

Knowledge Neither Begins Nor Ends in the Classroom

- We build on top of what you have already learned in this program and even high school.
- You are expected to put in the work outside the classroom
 - Attend to your pre-class work and readings
 - Come to office hours.
 - Experiment with code and parameters for fun!
- You are graduate students. Expected to take responsibility for learning!

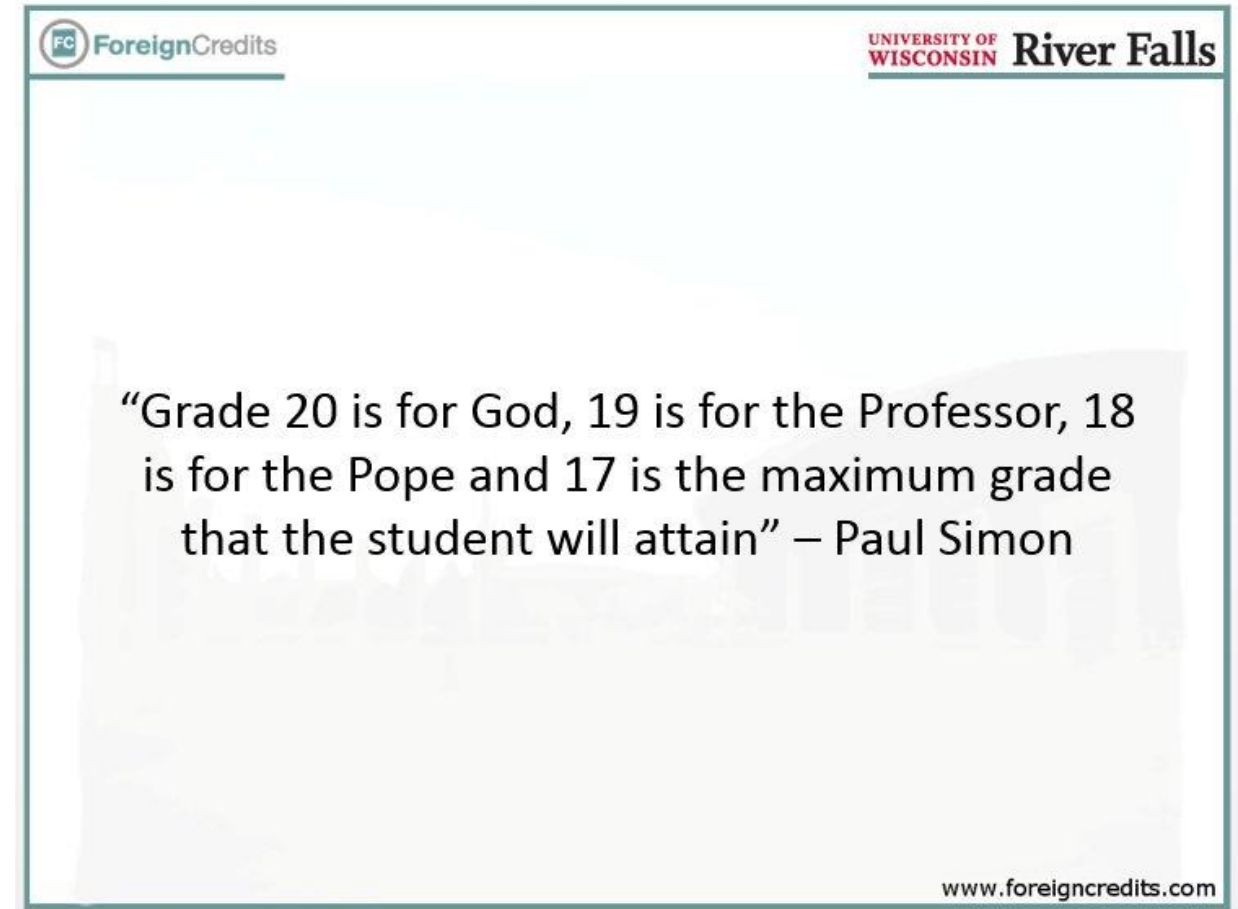


No Question is Foolish



Can I get an A?

- [Questrom policy](#)
- But you will get your fair chance.



Participation is Essential!

- 25% of your grade.
- Expectation:
 - Explaining, defending, or countering a position *with sound reasoning*.
 - Discussing and connecting ideas.
 - Looking at a problem from different lenses.
 - It is *NOT* about being correct.



Office Hours

- Meaning: “I did not understand” is not an excuse!
- A project-related question? Sync with your teammates in advance.
- Individual-work question? Do your homework and come with specific questions and proposed pathways.
 - A simple “How do I solve this?” is not a valid question.



Pass through the Syllabus

- Have you actually read it?!
- Pay attention to Piazza for announcements!
- No eating or drinking. Keep laptop and phone screens down unless instructed otherwise.
- **Zero Tolerance for Academic Integrity Violations!**

Intro to ML

What is Machine Learning?

- In the good old days (maybe even now...): An expert would come in and declare the golden rule...
- “If fever + cough, it’s an infection!”
- “In-store sales always spike near holidays.”
- “A cat has whiskers and pointy ears.”
- “Buy the dip!”

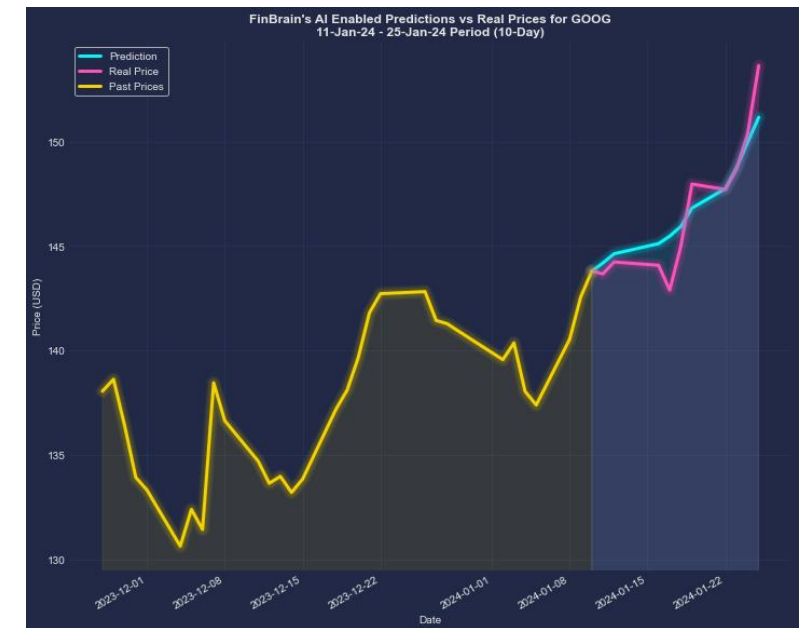


Knowledge-based modeling

What is Machine Learning?

- But then we decided to trust machines and data...
- Machines learn patterns we can't explicitly write down.
- They find structure that we may not notice.
- They adapt when the world changes.

<https://blog.finbrain.tech/>



Data-driven modeling

Data

- Is there enough of it?
- Does it need clean-up?



Damaged specimen

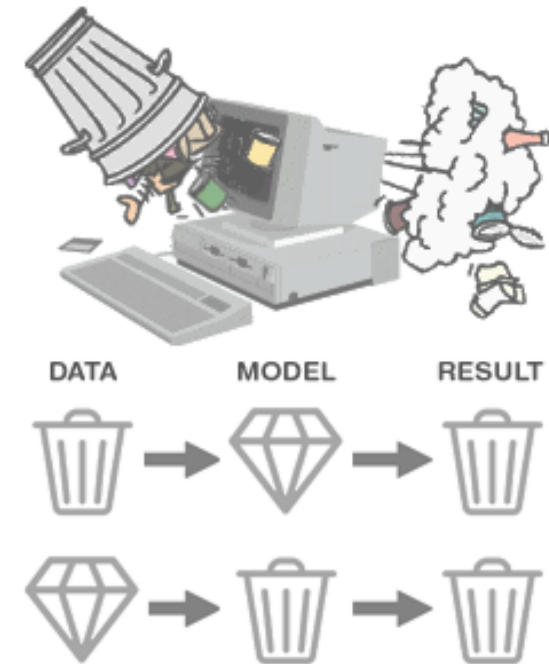


Missing Features



Occluded Features

Elhamod, M., Diamond, K. M., Murat Maga, A., Bakis, Y., Bart, H. L., Mabee, P., Dahdul, W., Leipzig, J., Greenberg, J., Avants, B., & Karpayne, A. (2022). Hierarchy-guided neural network for species classification. *Methods in Ecology and Evolution*, 13, 642–652. <https://doi.org/10.1111/2041-210X.13768>

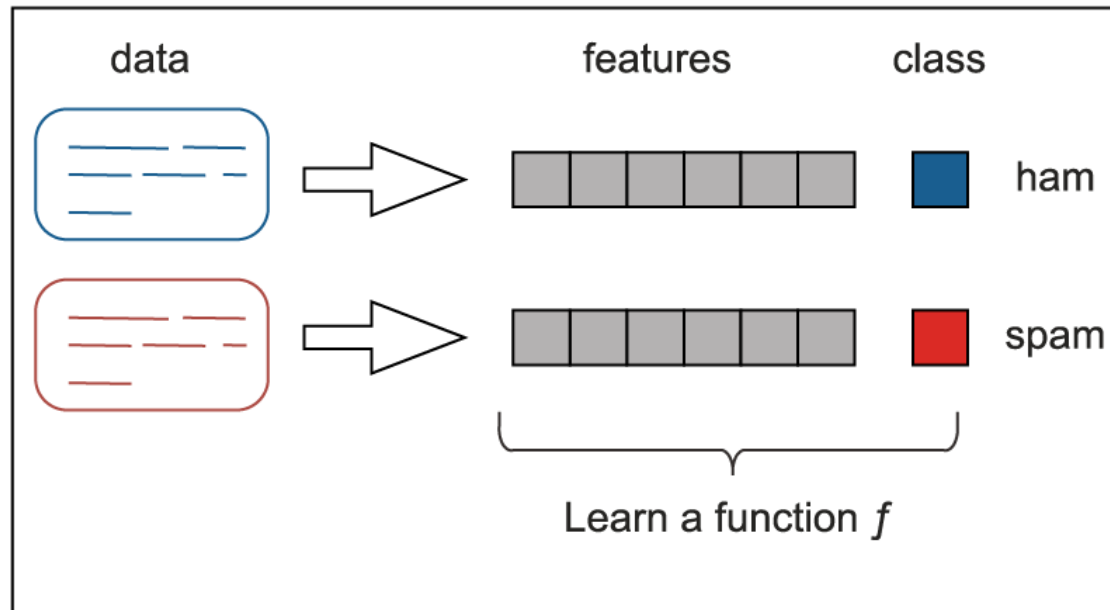


Resources

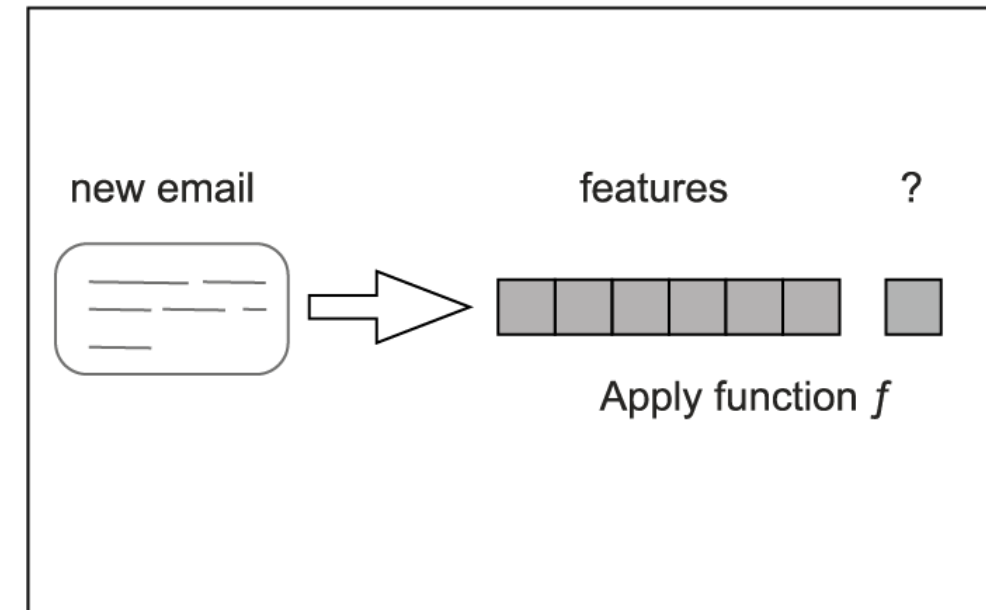
- What compute resources are available?
- What is the project's timeline?

How ML Works

Learning (training) phase



Prediction (testing) phase



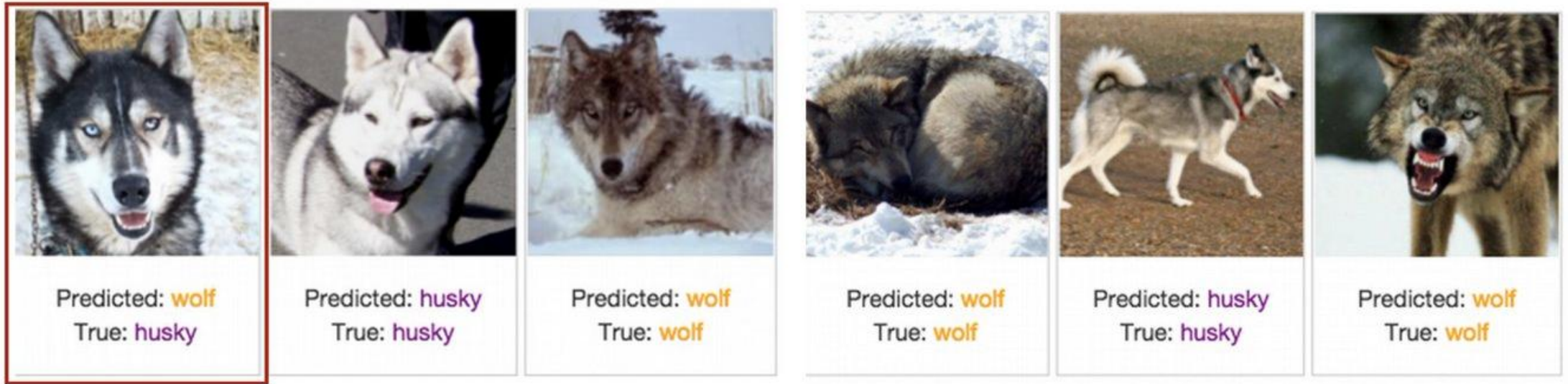
[Getting Started with NLP](#)

Intro to ML

Continued...

Model Generalization

- Example:



<https://medium.com/>

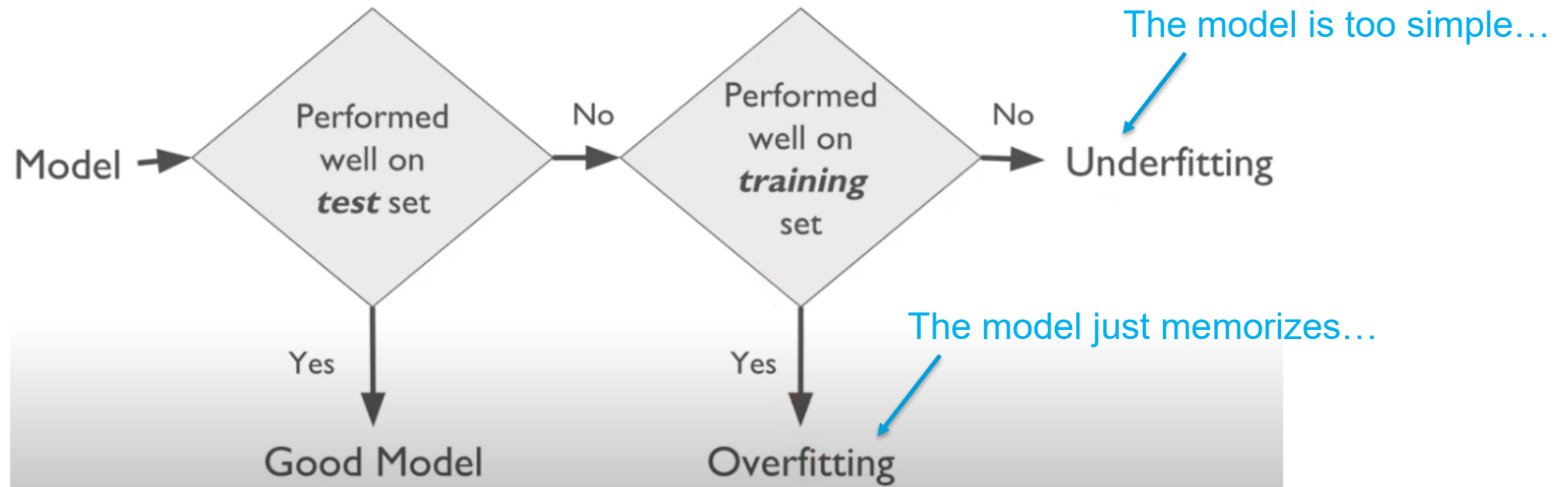
- Example 2

Model Generalization



<https://www.machinecurve.com>

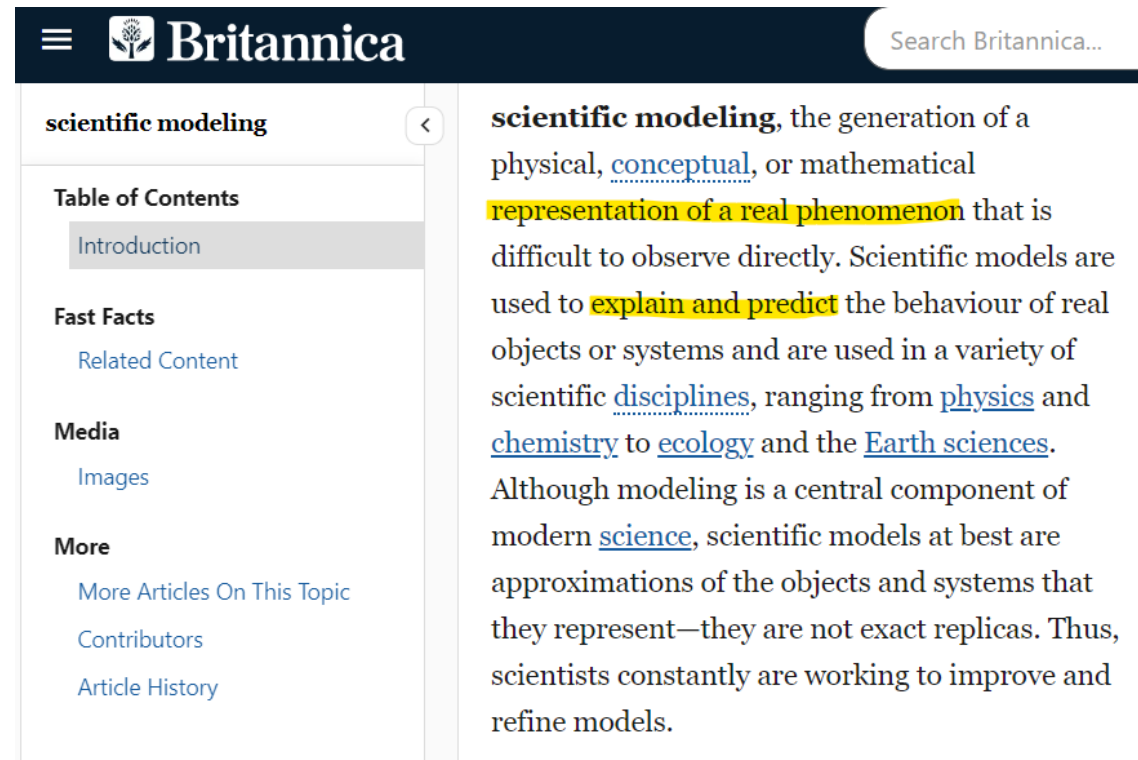
Overfitting and Underfitting



<http://jcsites.juniata.edu/faculty/rhodes/ml/clusterAn.htm>

Language Modeling

What is a model?



The screenshot shows the Britannica website interface. At the top, there is a dark blue header with the Britannica logo on the left and a search bar on the right. Below the header, the article title "scientific modeling" is displayed. To the left of the main text is a sidebar with a "Table of Contents" section where "Introduction" is highlighted. Other sidebar sections include "Fast Facts", "Media", and "More". The main content area on the right contains the introductory paragraph of the article, which defines scientific modeling and lists various scientific disciplines.

scientific modeling, the generation of a physical, [conceptual](#), or mathematical **representation of a real phenomenon** that is difficult to observe directly. Scientific models are used to **explain and predict** the behaviour of real objects or systems and are used in a variety of scientific [disciplines](#), ranging from [physics](#) and [chemistry](#) to [ecology](#) and the [Earth sciences](#). Although modeling is a central component of modern [science](#), scientific models at best are approximations of the objects and systems that they represent—they are not exact replicas. Thus, scientists constantly are working to improve and refine models.

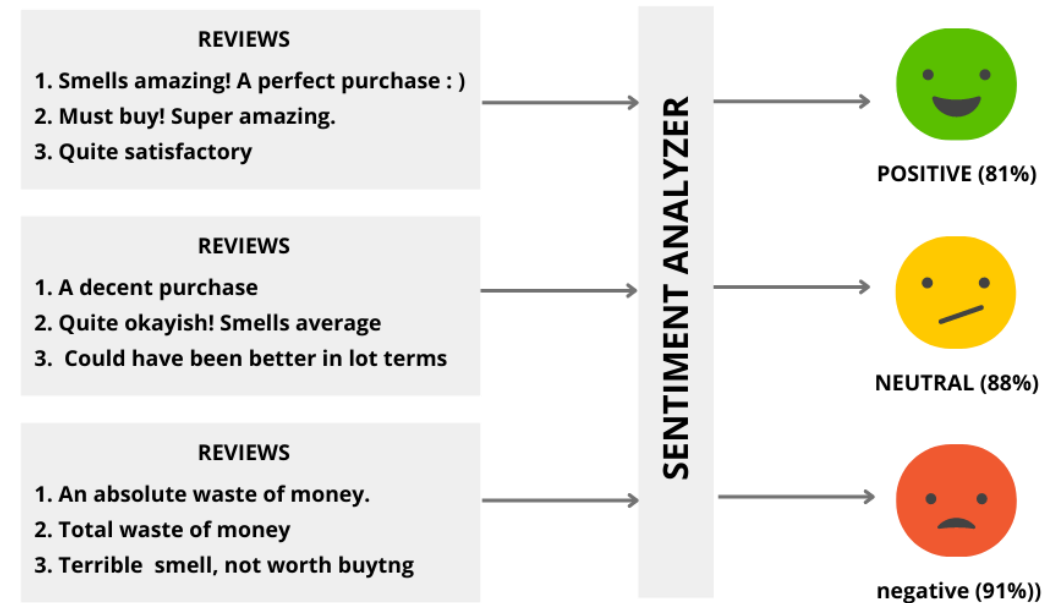
What is language modeling?...

Web search engine / ...

I saw a cat|

I saw a cat on the chair
I saw a cat running after a dog
I saw a cat in my dream
I saw a cat book

[Lena-volta](#)



What is language modeling?...

I grabbed the **branch** and broke it.

I went to the **branch** and deposited some money.

Context matters!

What is language modeling?...

- I went to ____.
- I woke up at 7 am and went to ____.
- I woke up at 7 am, packed my book and notebook, and went to ____.

The more context, the more certainty...

What is language modeling?...

I went to the **branch** and deposited some money.

I went to the **bank** and deposited some money.

I went to the **ATM** and deposited some money.

Words which frequently appear in **similar contexts** have **similar meaning**.

[Lena-voita](#)

Natural Language Processing (NLP)

Includes text generation:

- Text completion.
- Text summarization.
- Question answering.

But there are also many other tasks such as Text classification (e.g., Sentiment analysis, Reviews, Fake news) or word classification.

Formalizing our thoughts

- It seems we process language sequentially**.
- So, basic language modeling could be the chaining of word probabilities.

- How do we calculate these probabilities?

counting...

$P(\text{I saw a cat on } \dots) =$

$P(\text{I}) \cdot P(\text{saw}|\text{I}) \cdot P(\text{a}|\text{I saw}) \cdot P(\text{cat}|\text{I saw a}) \cdot P(\text{on}|\text{I saw a cat}) \cdot \dots$

Probability of I saw a cat on

$$P(\text{cat}) = \frac{N(\text{"cat" in corpus})}{N(\text{all words in corpus})}$$

$$P(\text{cat} | \text{my}) = \frac{N(\text{"my cat" in corpus})}{N(\text{"my" in corpus})}$$

Can you foresee any problem with this calculation?...

[Lena-volta](#)

N-grams

Instead, let's just use a context of ***fixed-length***.

$P(\text{I saw a cat on a mat}) =$

$P(\text{I})$

- $P(\text{saw} \mid \text{I})$
- $P(\text{a} \mid \text{I saw})$
- $P(\text{cat} \mid \text{I saw a})$
- $P(\text{on} \mid \text{I saw a cat})$
- $P(\text{a} \mid \text{I saw a cat on})$
- $P(\text{mat} \mid \text{I saw a cat on a})$

- $n=3$ (trigram model): $P(y_t | y_1, \dots, y_{t-1}) = P(y_t | y_{t-2}, y_{t-1})$,
- $n=2$ (bigram model): $P(y_t | y_1, \dots, y_{t-1}) = P(y_t | y_{t-1})$,
- $n=1$ (unigram model): $P(y_t | y_1, \dots, y_{t-1}) = P(y_t)$.

[Lena-voita](#)

N-grams

Context is like a sliding window into the past.

Hugging Face is a startup based in New York City and Paris

p(word)

[Huggingface](#)

In-Class Work

Resources

- [What does it mean for computers to understand language? \(vcubingx, 12 min\)](#)
- [Meet Moltbook, the social site where AI agents run wild](#)
- [Math teachers stage a calculated protest](#)
- [It's time to believe in the AI hype](#)
- [Did GPT4 really pass the bar exam?](#)
- [Amazon “AI” stores.](#)
- [Human oversight of Robotaxis.](#)
- [Gemini demo was faked.](#)
- [Meaning and calculation of perplexity.](#)
- [AI hype is completely out of control](#)
- [No, AI is not like the DotCom bubble](#)